

# Fishing for Sustainability

## Activity Procedure

Fri Sep 11, 2026

### Introduction

In this activity, you will simulate a **shared fishery**—a resource that everyone can access but no one owns.

You should work in **groups of ~4-6 people** for the entire exercise.

You will participate in three short rounds, each with different rules. Your goal is to observe how:

- Individual decisions affect the **entire population**
- Different management strategies influence **sustainability**
- Human behavior creates challenges in managing **common-pool resources**

### Your Role

You are an independent fisher.

- Your goal is to **maximize your personal catch**
- However, the fish population is **shared by everyone**
- Your decisions will affect both **your outcome** and the **group outcome**

### The System (Applies to All Rounds)

- The lake begins with **30 fish**
- After each round:
  - The remaining fish **reproduce (double)**
  - The population cannot exceed **30 fish** (carrying capacity,  $K$ )

## What Changes Each Round

| Round | System       | Key Idea                                      |
|-------|--------------|-----------------------------------------------|
| 1     | Open Access  | No rules or communication                     |
| 2     | Fixed Quota  | Group agrees on a total limit                 |
| 3     | Fixed Effort | Each person follows the same individual limit |

## Key Concepts

You should be able to apply these ideas after the activity:

**Tragedy of the Commons** When individuals acting in their own interest deplete a shared resource

**Social Dilemma** When what is best for individuals conflicts with what is best for the group

**Carrying Capacity (K)** The maximum population size the environment can sustain (here, 30 fish)

**Maximum Sustainable Yield (MSY)** The largest long-term harvest that does not reduce the population over time

## Round 1: Open Access

Test whether a **per-person rule** improves sustainability, fairness, and compliance.

You are an independent fisher trying to **maximize your personal catch**. Unlike Round 2, this system sets a **strict individual limit** that everyone must follow.

- 1 bowl = **lake** (starts with **30 “fish”**)
- 8 cups = **boats** (one per student)

### ! Biological Rule

After harvest, fish remaining in the lake **double** (up to a **maximum of 30**).

If the fishery **collapses**, no fish remain and **no one gains anything further from this round**.

## Communication Rules

Read these aloud & follow exactly

- **Do NOT talk**, signal, make eye contact, or coordinate in any way.
- **You are independent fishers**. No communication is permitted **until after** this handout is collected.

### ! You may communicate:

Nothing at all. No talking, signaling, or writing to each other.

**Reminder:** Some fishers may try to maximize their catch. Others may try to conserve. You do **not** know what others will do.

## Procedure

- (1) **Secret Choice (0–6):** Privately decide how many fish to take (0–6). Write the number on your paper; **do not show** anyone.
- (2) **Simultaneous Harvest:** When the instructor says “**Fish!**” everyone takes their chosen fish at the same time.
- (3) **If total fish taken > fish available:** The fishery **collapses** (0 fish remain). **No recovery. Stop.**
- (4) **If fish remain:** Instructor counts the remaining fish, then **doubles** them (up to 30).

- (5) **Record** your results in the data table for this round and answer the reflection questions **before** you proceed.

## Round 2: Fixed Quota

Test whether a **group quota** improves sustainability—and whether people will follow it.

You are an independent fisher trying to **maximize your personal catch**. The class may agree on a quota, but your **individual payoff still depends on how many fish you take**.

- Reset lake to **30 fish**
- Same 8 boats (cups)

### ! Biological Rule

After harvest, fish remaining in the lake **double** (up to a **maximum of 30**).  
If the class exceeds the quota, the fishery **collapses** (0 fish remain).

**There is no recovery for this round.**

## Communication Rules

Read these aloud & follow exactly

- **Discussion Allowed (2 minutes ONLY):** You may talk to agree on a **total class quota** (e.g., “We will catch at most 16 fish”).
- After the 2-minute discussion ends:
  - **No further communication** (no talking, signaling, or eye contact)
  - All decisions are **private**

### ⚠ You may communicate (for 2 minutes only):

- Propose and agree on a **single total class quota**

### You may NOT communicate (after time ends):

- Your personal intended catch
- Any signals or coordination during harvesting

**Reminder:** The quota is a **group agreement**, not automatically enforced. Individuals can still choose how much to take.

## Procedure

- (6) **Set the Quota (group, 2 minutes):** Agree on a total class quota: **Class quota =** \_\_\_\_\_ **fish**
- (7) **Secret Choice (0–6):** After discussion ends, privately choose how many fish to take (0–6). **Do not reveal your number.**
- (8) **Simultaneous Harvest:** When the instructor says “**Fish!**” everyone takes their chosen fish at the same time.
- (9) **Check Outcome:**
  - If **total class catch**  $>$  **quota**  $\rightarrow$  fishery **collapses** (0 fish remain). **Stop.**
  - If **total class catch**  $\leq$  **quota**  $\rightarrow$  instructor counts remaining fish, then **doubles** them (up to 30).
- (10) **Record** your results in the data table for this round and answer the reflection questions **before** you proceed.

### Round 3: Fixed Effort

Test whether a **per-person rule** improves sustainability, fairness, and compliance.

You are an independent fisher trying to **maximize your personal catch**. Unlike Round 2, this system sets a **strict individual limit** that everyone must follow.

- Reset lake to **30 fish**
- Same 8 boats (cups)

#### ! Biological Rule

After harvest, fish remaining in the lake **double** (up to a **maximum of 30**).

#### Key Difference from Round 2:

The rule applies **equally to each individual** and is **fully enforced**.

### Communication Rules

Read these aloud & follow exactly

- **Discussion Allowed (2 minutes):** You may talk to agree on a **per-person limit** (e.g., “2 fish per fisher”).
- After the 2-minute discussion ends:
  - **No further communication** (no talking, signaling, or eye contact)
  - Everyone must follow the agreed rule exactly

#### ! You may communicate (for 2 minutes only):

- Propose and agree on a **per-person limit**

#### You must follow:

- Take **exactly** the agreed number of fish
- Do **not** exceed or go below the limit

#### You may NOT:

- Signal, negotiate, or change behavior during harvesting

**Reminder:** This rule is **enforceable**—everyone is held to the same limit. You cannot increase your payoff by taking more than others.

## Procedure

- (11) **Set the Per-Person Limit (group, 2 minutes):** Per-person limit = \_\_\_\_\_ fish
- (12) **Harvest Exactly the Limit:** When the instructor says “**Fish!**” each fisher takes **exactly** the agreed limit.
- (13) **Reproduction:** Instructor counts remaining fish, then **doubles** them (up to 30).
- (14) **Record** your results in the data table for this round and answer the reflection questions.